

SHLS - Shoals Technologies Group | Full Deep Dive

September 21, 2026 | Framework 6.0.2 | NOT HELD | STRUCTURAL - power interconnection; TEMPORAL - solar project and factory cycle | Watchlist

Analytical Base: 42/60 (70%). Valuation: 20x forward four-quarter adjusted operating profit (adjusted EBITDA less ordinary depreciation).

Snapshot: USD 7.60 per share on September 18, 2026 close; 168.288 million shares, implying USD 1,278.99 million equity value. Source [SHLS-5]. Price is a dated header snapshot, not a valuation input in the body.

Active horizon Q3 2027 ceiling: USD 21.35/share. Q1 2028: USD 23.69/share.

All financial model amounts are USD millions, shares in millions, unless stated. Calendar-year reporting. All forecasts use calendar quarters; new-product sales are probability-weighted once in the operating model.

1. Thesis, bottleneck and expanded TAM

Shoals makes electrical balance-of-system equipment that connects generation and storage assets. The Framework 6.0 link is power delivery: more solar and battery capacity requires reliable electrical connections and cost-effective installation. Its wiring architecture and project engineering can reduce labor and execution risk. This is a qualified industrial supplier, not ownership of all electricity or data-center spending.

The investment case combines a recovering core with two different adjacencies. Battery energy storage systems (BESS) already contribute sales. Air Link, a proposed rack-power architecture, remains a qualification opportunity. The latter must earn its way into revenue through testing, safety approvals and customer adoption. Neither a memorandum nor an engineering discussion is a purchase order. [SHLS-2]

Manufacturing consolidation can improve productivity, but weak gross margin despite strong revenue shows why factory output and product quality are central. The model gives credit for gradual recovery rather than immediately restoring historical peak margins.

The 2025 annual report identifies Construction Innovation, Hikam America, Nextpower/Bentek, Premier PV, TerraSmart and Voltage as competitors. Installation cost, reliability, system design and warranty support matter as much as component price. Patents can defend a design while litigation consumes cash; customer concentration and project bidding prevent treating this as a monopoly. [SHLS-4]

TAM expansion and S-curve

Phase	Economic exposure	Model treatment
Phase 1 - current, 100% relevance	Utility-scale solar EBOS and existing BESS installations	Core solar/OEM and shipping storage sales
Phase 2 - 2027-2028, 50% conceptual weight	Larger storage systems and additional EPC relationships	Explicit conversion within BESS revenue
Phase 3 - 2028 onward, 20% conceptual weight	Rack power via Air Link and international expansion	Small probability-weighted pilot-to-production path; no separate option value

No sufficiently current, comparable dollar TAM was verified for Shoals-specific content across all three markets. Total solar generation or data-center capex would materially overstate its addressable sales. A bottom-up commercial scenario is therefore used: TerraFlow discussed 5GW of annual storage potential, but this is an MOU, not contracted revenue. At an assumed \$10M of Shoals content per GW, the full scenario would be \$50M annually. The \$5-15M/GW sensitivity is an analyst proxy with LOW confidence, not a disclosed ASP. [SHLS-2]

2. Latest evidence and financial anchor

Q2 revenue was \$163.372M, GAAP gross profit \$49.525M and OP \$18.722M. Adjusted EBITDA was \$31.550M. Backlog plus awarded orders totaled \$801.4M; awarded orders are not signed commitments. FY2026 guidance remains revenue \$600-640M and adjusted EBITDA \$118-132M; Q3 guidance is \$150-170M and \$32-37M. [SHLS-1]

Two-call review: May 5 described factory-transition pressure and a larger opportunity book. August 4 reported \$20M quarterly BESS sales, largely associated with ON.energy, and \$699.7M of backlog/awards scheduled over the following four quarters. Shipment timing and final contracts still matter. [SHLS-2, SHLS-3]

The August call places TerraFlow revenue in 2027, not 2026. Air Link was approaching live testing with pricing still under development. Q4 has fewer production days, while reduced IP litigation expense may help profit. The forecast therefore allows Q4 sales to decline sequentially even as profitability improves. [SHLS-2]

Period	Status	Revenue	Gross profit	Operating profit	Source
Q1 2026	Actual GAAP	140.56	41.01	7.72	SHLS-1 H1 minus Q2
Q2 2026	Actual GAAP	163.37	49.52	18.72	SHLS-1

For comparability with the framework OP convention, adjusted EBITDA is reduced by ordinary depreciation. Q2 therefore implies \$28.810M adjusted OP (\$31.550M minus \$2.740M); this is an analyst-derived metric. Intangible amortization, SBC and company-specified exceptional items remain excluded. It is not GAAP OP. H1 adjusted EBITDA of \$52.662M anchors the full-year bridge. [SHLS-1]

The forecast uses \$620M FY2026 revenue and \$125M adjusted EBITDA, both guidance midpoints. After actual H1 revenue of \$303.929M, Q3 is \$160M and Q4 is \$156.071M. Q3 EBITDA is \$34.5M, and Q4 is the residual \$37.838M. Assumed depreciation of \$3.0M/\$3.2M yields modeled OP of \$31.5M/\$34.638M. The SG&A allocation is solved to this bridge, rather than imposing an incompatible margin target.

Balance sheet and capital allocation

June cash was \$15.724M versus \$196.750M on the revolver; inventory reached \$184.720M. The large inventory build means earnings recovery must translate into cash. FY2026 operating cash flow guidance is \$65-85M, capex \$20-30M and interest \$8-12M. [SHLS-1] Patent defense, warranty remediation and project delays can absorb otherwise distributable cash; this model does not assume immediate debt elimination or large buybacks.

3. Base score and evidence ledger

Category	Score	Evidence
Core Thesis	15	Qualified electrical systems and installation advantage; competitive bidding
Cycle Stage	8	Growing orders with factory stabilization still underway
TAM Expansion	8	Storage adjacency proven; rack power not yet commercially proven
Catalysts	4	Productivity recovery, 2027 storage conversion and Air Link testing
Macro	8	Power availability and energy investment; policy and financing sensitivity
Alpha	3	Separating signed demand, awards and unqualified options
Risks	-4	Execution/quality, project policy, competition/legal, liquidity
Total	42/60 = 70%	Watchlist

42/60 = 70, Watchlist. Storage contributes to conviction; unqualified rack power does not receive the same evidence weight.

4. Quarterly operating model

Three vectors expose different maturity and margins. Solar/core includes OEM and other established products; storage and Air Link are modeled separately to prevent double counting. Gross margins are analytical allocations and include ordinary depreciation. R&D is separated from other operating expense for transparency although company reporting aggregates functions differently.

Revenue vector	Driver and source	Confidence
Solar EBOS, OEM and established products	SHLS-1/3 order book; delivery and pricing proxy	MEDIUM near term; LOW beyond 2027
Battery storage interconnection	SHLS-2 shipping ON.energy demand; probability-weighted new projects	MEDIUM existing sales; LOW new-customer ramps
Air Link rack power	SHLS-2 development disclosure; analyst adoption scenario	LOW - pre-commercial qualification

Solar/core growth reflects project conversion and market participation, not a forecast of rising copper prices. BESS includes existing shipments plus new customer adoption. TerraFlow contributes an illustrative \$7.5M in 2027: 5GW x \$10M/GW x 30% utilization x 50% conversion probability. It is inside the \$115M BESS total, not added twice. The rest is \$80M existing/repeat demand and \$27.5M other project wins. Air Link revenue assumes \$10M/\$50M/\$110M/\$180M of potential 2027-2030 programs at 50% adoption probability; resulting modeled sales are \$5M/\$25M/\$55M/\$90M. No further probability discount is applied to the ceiling. These are revenue-envelope proxies because reliable rack quantities and ASPs are unavailable.

Physical and commercial drivers

Calendar year	Solar/core	BESS	Air Link after 50% weighting	Total sales	
2027		625	115	5	745
2028		690	165	25	880
2029		750	220	55	1,025
2030		810	285	90	1,185

Vector revenue, gross margin and gross profit

For each row: vector gross profit = vector revenue x its own gross margin. The following tables expose the inputs; they are analyst model assumptions unless explicitly identified as reported or guided.

Solar EBOS, OEM and established products

Quarter	Revenue	Gross margin	Gross profit
Q3 2026		135.00	33.50% 45.23
Q4 2026		134.07	35.30% 47.33
Q1 2027		143.75	36.00% 51.75
Q2 2027		156.25	36.00% 56.25
Q3 2027		168.75	36.00% 60.75
Q4 2027		156.25	36.00% 56.25
Q1 2028		158.70	36.50% 57.93
Q2 2028		172.50	36.50% 62.96
Q3 2028		186.30	36.50% 68.00
Q4 2028		172.50	36.50% 62.96
Q1 2029		172.50	37.00% 63.82
Q2 2029		187.50	37.00% 69.38
Q3 2029		202.50	37.00% 74.92
Q4 2029		187.50	37.00% 69.38
Q1 2030		186.30	37.00% 68.93
Q2 2030		202.50	37.00% 74.92
Q3 2030		218.70	37.00% 80.92
Q4 2030		202.50	37.00% 74.92

Battery storage interconnection

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	25.00	30.00%	7.50
Q4 2026	22.00	32.00%	7.04
Q1 2027	26.45	33.00%	8.73
Q2 2027	28.75	33.00%	9.49
Q3 2027	31.05	33.00%	10.25
Q4 2027	28.75	33.00%	9.49
Q1 2028	37.95	34.00%	12.90
Q2 2028	41.25	34.00%	14.03
Q3 2028	44.55	34.00%	15.15
Q4 2028	41.25	34.00%	14.03
Q1 2029	50.60	35.00%	17.71
Q2 2029	55.00	35.00%	19.25
Q3 2029	59.40	35.00%	20.79
Q4 2029	55.00	35.00%	19.25
Q1 2030	65.55	35.00%	22.94
Q2 2030	71.25	35.00%	24.94
Q3 2030	76.95	35.00%	26.93
Q4 2030	71.25	35.00%	24.94

Air Link rack power

Quarter	Revenue	Gross margin	Gross profit	
Q3 2026		0.00	0.00%	0.00
Q4 2026		0.00	0.00%	0.00
Q1 2027		1.15	25.00%	0.29
Q2 2027		1.25	25.00%	0.31
Q3 2027		1.35	25.00%	0.34
Q4 2027		1.25	25.00%	0.31
Q1 2028		5.75	38.00%	2.19
Q2 2028		6.25	38.00%	2.38
Q3 2028		6.75	38.00%	2.56
Q4 2028		6.25	38.00%	2.38
Q1 2029		12.65	42.00%	5.31
Q2 2029		13.75	42.00%	5.77
Q3 2029		14.85	42.00%	6.24
Q4 2029		13.75	42.00%	5.77
Q1 2030		20.70	44.00%	9.11
Q2 2030		22.50	44.00%	9.90
Q3 2030		24.30	44.00%	10.69
Q4 2030		22.50	44.00%	9.90

Combined quarterly earnings

Quarter	Status	Revenue	Gross profit	R&D	SG&A	Other opex	OP	OP margin
Q3 2026	Model	160.00	52.73	4.00	17.23	0.00	31.50	19.69%
Q4 2026	Model	156.07	54.37	3.90	15.83	0.00	34.64	22.19%
Q1 2027	Model	171.35	60.77	4.28	18.85	0.00	37.63	21.96%
Q2 2027	Model	186.25	66.05	4.66	20.49	0.00	40.91	21.96%
Q3 2027	Model	201.15	71.33	5.03	22.13	0.00	44.18	21.96%
Q4 2027	Model	186.25	66.05	4.66	20.49	0.00	40.91	21.96%
Q1 2028	Model	202.40	73.01	5.06	21.25	0.00	46.70	23.07%
Q2 2028	Model	220.00	79.36	5.50	23.10	0.00	50.76	23.07%
Q3 2028	Model	237.60	85.71	5.94	24.95	0.00	54.82	23.07%
Q4 2028	Model	220.00	79.36	5.50	23.10	0.00	50.76	23.07%
Q1 2029	Model	235.75	86.85	5.89	23.58	0.00	57.38	24.34%
Q2 2029	Model	256.25	94.40	6.41	25.62	0.00	62.37	24.34%
Q3 2029	Model	276.75	101.95	6.92	27.68	0.00	67.36	24.34%
Q4 2029	Model	256.25	94.40	6.41	25.62	0.00	62.37	24.34%
Q1 2030	Model	272.55	100.98	6.81	26.71	0.00	67.46	24.75%
Q2 2030	Model	296.25	109.76	7.41	29.03	0.00	73.32	24.75%
Q3 2030	Model	319.95	118.54	8.00	31.36	0.00	79.19	24.75%
Q4 2030	Model	296.25	109.76	7.41	29.03	0.00	73.32	24.75%

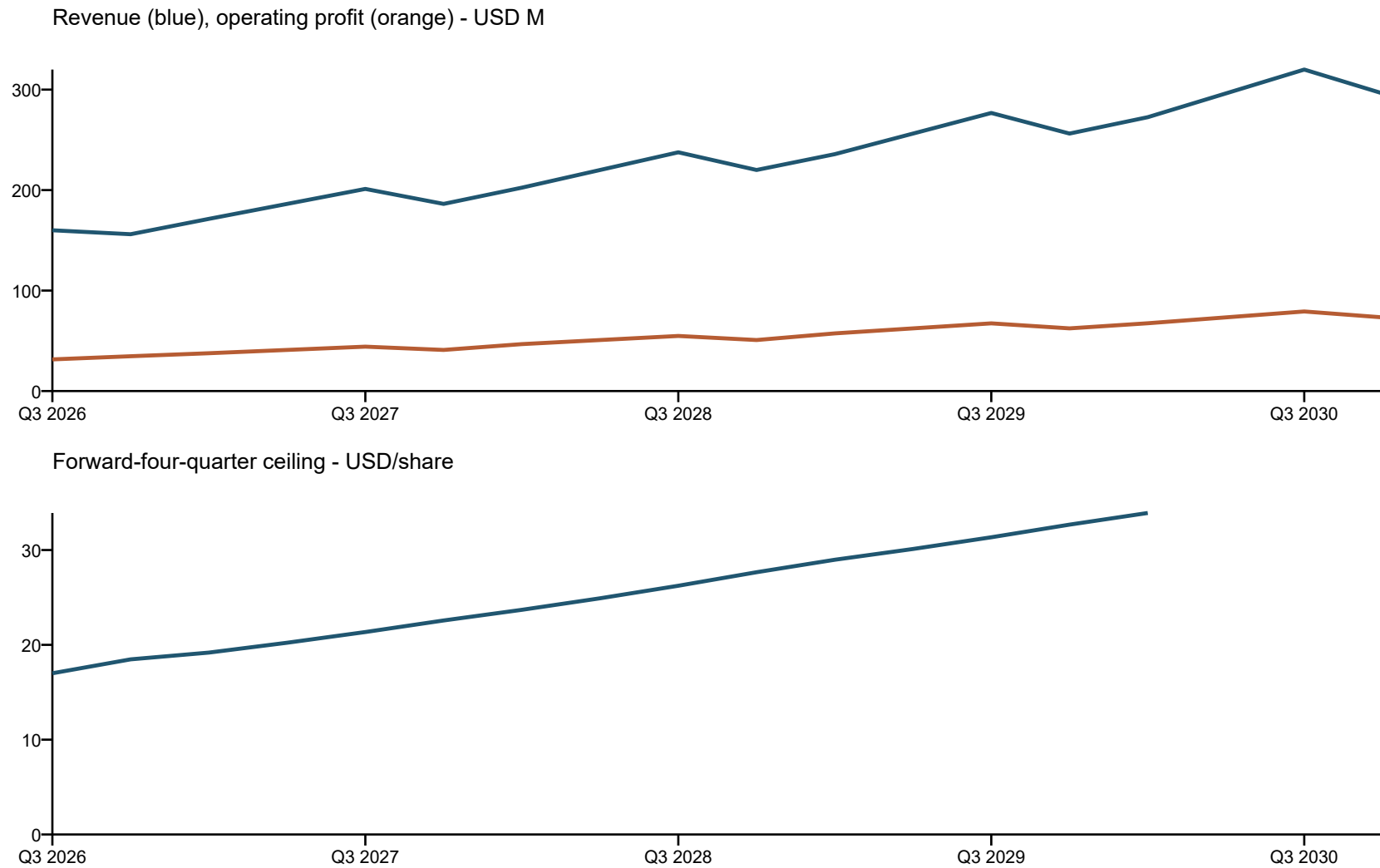
Annual trajectory cross-check

The calendar-bucket sums below are a capacity and demand cross-check; quarterly windows remain the live valuation method. Years beyond the latest guidance are model extensions, not company guidance.

Year	Revenue	Operating profit	OP margin
2027	745.00	163.62	21.96%
2028	880.00	203.05	23.07%
2029	1,025.00	249.48	24.34%
2030	1,185.00	293.30	24.75%

The model grows sales from \$620M in 2026 to \$745M in 2027, then \$1,185M in 2030. This requires approximately 17.6% annual growth over 2026-2030 and substantial storage diversification. It is an analyst case, not long-term guidance. A \$20M BESS shortfall at 33% gross margin removes \$6.6M GP before expense flexibility. Delaying Air Link to 2029 removes \$5M/\$25M sales from 2027/2028. Backlog/awards are a timing cross-check rather than an extra revenue vector.

5. Forward four-quarter ceiling and derivations



20x forward-four-quarter adjusted operating profit (adjusted EBITDA less ordinary depreciation) is the portfolio owner-selected multiple as of September 21, 2026, held constant across the model. This is the framework operating-profit convention, not an observed market P/E. A one-turn change changes each ceiling by 5%. Adjusted OP remains adjusted EBITDA less ordinary depreciation. Keeping \$4.4M quarterly SBC as an economic expense would reduce annualized OP by about \$17.6M.

The framework convention applies the operating-profit multiple directly to diluted equity shares. It is neither a net-income P/E nor an enterprise-value bridge. No separate net-cash addition, tax conversion, probability haircut or price-dependent ranking enters this ceiling. Any probability-weighted revenue is already included once in its operating vector.

Ceiling quarter	Exact earnings window	Forward-4Q OP	Multiple	Diluted shares	Ceiling/share
Q3 2026	Q3 2026 to Q2 2027	144.68	20x	170.200	17.00
Q4 2026	Q4 2026 to Q3 2027	157.36	20x	170.400	18.47
Q1 2027	Q1 2027 to Q4 2027	163.62	20x	170.600	19.18
Q2 2027	Q2 2027 to Q1 2028	172.69	20x	170.800	20.22
Q3 2027	Q3 2027 to Q2 2028	182.55	20x	171.000	21.35
Q4 2027	Q4 2027 to Q3 2028	193.19	20x	171.200	22.57
Q1 2028	Q1 2028 to Q4 2028	203.05	20x	171.400	23.69
Q2 2028	Q2 2028 to Q1 2029	213.73	20x	171.600	24.91
Q3 2028	Q3 2028 to Q2 2029	225.33	20x	171.800	26.23
Q4 2028	Q4 2028 to Q3 2029	237.87	20x	172.000	27.66
Q1 2029	Q1 2029 to Q4 2029	249.48	20x	172.200	28.98
Q2 2029	Q2 2029 to Q1 2030	259.55	20x	172.400	30.11
Q3 2029	Q3 2029 to Q2 2030	270.51	20x	172.600	31.35
Q4 2029	Q4 2029 to Q3 2030	282.34	20x	172.800	32.68
Q1 2030	Q1 2030 to Q4 2030	293.30	20x	173.000	33.91

Complete derivation for every ceiling quarter

Q3 2026: OP window Q3 2026 to Q2 2027: $(31.50 + 34.64 + 37.63 + 40.91) \times 20 / 170.200 = \text{USD } 17.00$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2026: OP window Q4 2026 to Q3 2027: $(34.64 + 37.63 + 40.91 + 44.18) \times 20 / 170.400 = \text{USD } 18.47$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2027: OP window Q1 2027 to Q4 2027: $(37.63 + 40.91 + 44.18 + 40.91) \times 20 / 170.600 = \text{USD } 19.18$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2027: OP window Q2 2027 to Q1 2028: $(40.91 + 44.18 + 40.91 + 46.70) \times 20 / 170.800 = \text{USD } 20.22$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2027: OP window Q3 2027 to Q2 2028: $(44.18 + 40.91 + 46.70 + 50.76) \times 20 / 171.000 = \text{USD } 21.35$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2027: OP window Q4 2027 to Q3 2028: $(40.91 + 46.70 + 50.76 + 54.82) \times 20 / 171.200 = \text{USD } 22.57$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2028: OP window Q1 2028 to Q4 2028: $(46.70 + 50.76 + 54.82 + 50.76) \times 20 / 171.400 = \text{USD } 23.69$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2028: OP window Q2 2028 to Q1 2029: $(50.76 + 54.82 + 50.76 + 57.38) \times 20 / 171.600 = \text{USD } 24.91$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2028: OP window Q3 2028 to Q2 2029: $(54.82 + 50.76 + 57.38 + 62.37) \times 20 / 171.800 = \text{USD } 26.23$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2028: OP window Q4 2028 to Q3 2029: $(50.76 + 57.38 + 62.37 + 67.36) \times 20 / 172.000 = \text{USD } 27.66$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2029: OP window Q1 2029 to Q4 2029: $(57.38 + 62.37 + 67.36 + 62.37) \times 20 / 172.200 = \text{USD } 28.98$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2029: OP window Q2 2029 to Q1 2030: $(62.37 + 67.36 + 62.37 + 67.46) \times 20 / 172.400 = \text{USD } 30.11$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2029: OP window Q3 2029 to Q2 2030: $(67.36 + 62.37 + 67.46 + 73.32) \times 20 / 172.600 = \text{USD } 31.35$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2029: OP window Q4 2029 to Q3 2030: $(62.37 + 67.46 + 73.32 + 79.19) \times 20 / 172.800 = \text{USD } 32.68$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2030: OP window Q1 2030 to Q4 2030: $(67.46 + 73.32 + 79.19 + 73.32) \times 20 / 173.000 = \text{USD } 33.91$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

The final three operating quarters support earlier four-quarter windows and have no standalone ceiling. Arithmetic uses unrounded inputs; displayed components are rounded for readability.

The model starts at 170.2M diluted shares against 168.288M basic outstanding and 170.023M latest diluted weighted average. It adds 0.2M each quarter for net compensation dilution. No buyback is assumed while liquidity and working capital remain important. [SHLS-1]

Mechanical share sensitivity

Q1 2028 denominator	Ceiling/share
Current model	23.69
5% more shares	22.56
10% more shares	21.54

6. Cross-book relationships and monitoring

FSLR sells generation modules; Shoals sells installation and electrical infrastructure. The same solar project can benefit both, creating common project-policy exposure without identical revenue.

EnerSys provides batteries and power systems. Shoals can benefit from storage deployment but a portfolio-level link does not establish a supplier contract between these companies.

AI data-center growth supports power infrastructure broadly. Optical TAM forecasts and memory pricing are not inputs to Shoals revenue; Air Link requires its own customer evidence.

7. Scored risks and re-rate tests

Risk class	Score	Failure channel
Factory and quality execution	-1	Rework, warranty and ramp inefficiencies prevent margin recovery
Project finance and policy	-1	Customer financing, permitting or incentive changes defer shipments
Competition and intellectual property	-1	Aggressive bids or unfavorable legal outcomes weaken differentiation and cost recovery
Liquidity and concentration	-1	Inventory, debt and large project customers amplify cash-flow volatility

Dated triggers and counters

Checkpoint	Positive evidence	Counter / model action
Q3 2026 results	Revenue and EBITDA meet guide; inventory conversion improves	Margin recovery fails despite higher throughput
Q4 2026 results	Factory productivity and lower legal spending improve OP	Exceptional costs recur as normal economics
2027 storage conversion	TerraFlow moves from MOU to funded purchase orders	Do not book full 5GW potential as Shoals revenue
2027-2028 Air Link gates	Testing, approvals, published pricing and initial repeat orders	Defer the option if customer qualification remains incomplete

Financial actuals and current guidance are high-confidence as disclosed. Gross-margin allocation, future BESS content and Air Link probability are analyst assumptions. There is no verified current numeric total market estimate for the precise product scope. Plan B commercial envelopes avoid pretending that broad electricity investment is Shoals-addressable sales.

8. Sources and snapshot

- **SHLS-1** - August 4, 2026. Q2 results and financial reconciliations. Actuals, share count and guide; exact H1 EBITDA bridge.
- **SHLS-2** - August 4, 2026. Q2 management call. Management remarks: BESS, TerraFlow, Air Link and shipment cadence.
- **SHLS-3** - May 5, 2026. Q1 management call. Previous-quarter factory and order context.
- **SHLS-4** - 2025 annual report. Annual filing. Competition, customer concentration, IP and manufacturing.
- **SHLS-5** - September 18, 2026. Historical closing quote. Dated header only.

Model snapshot: September 21, 2026. SHLS. Framework 6.0.2. Base 42/60 (70%). Eighteen modeled operating quarters and fifteen complete forward-four-quarter ceilings. Current full PDF and synchronized Markdown: G:/My Drive/1amInvesting/Artifacts/SHLS_Shoals_DeepDive.pdf and .md. The companion company worksheet uses this same model and links the ceiling path to the master portfolio.